Reproducing Matcher: Training-Free One-Shot Segmentation

Bidirectional Matching as a Robust Outlier Filter for DINOv2 + SAM

Jiyao Xi Bingling Liu Jingzei Dai

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Problem & Statistical Setup

One-shot semantic segmentation: given a single labeled *reference* image (image + binary mask) and an unlabeled *target* image, predict the binary segmentation mask of the same concept on the target — **without training any model**.

Formally, for reference (x_r, m_r) and target x_t, predict $\hat{m}_t \in \{0,1\}^{H\times W}$. Performance measured by intersection-over-union against ground-truth m_t:

PER-EPISODE METRIC

$$\mathrm{IoU} = |\hat{m}_t \cap m_t| / |\hat{m}_t \cup m_t|$$

Estimand of interest is the population-level mean IoU:

ESTIMAND

$$\mathrm{mIoU} = \mathbb{E}_{(x_r,x_t)\sim\mathcal{D}_c}[\,\mathrm{IoU}(\hat{m}_t, m_t)\,]$$

where $\mathcal{D}_c$ is the joint distribution of (ref, target) pairs from class c. Estimated via Monte Carlo over $N = 2{,}400$ episodes (20 classes $\times$ 120 (ref, target) pairs / class).

Two regimes compared: ϕ_{full} (with bidirectional matching, paper Table 1) vs. ϕ_{fwd} (forward-only, paper Table 4b). Effect: $\Delta = \mathrm{mIoU}(\phi_{\text{full}}) - \mathrm{mIoU}(\phi_{\text{fwd}})$.

Key Assumption: Feature Geometry

The whole pipeline rests on a property of DINOv2: **semantic similarity $\approx$ cosine similarity in feature space**, used for patches encoding across images,

WORKING ASSUMPTION

$$\langle z_r, z_t\rangle / (\|z_r\|\cdot\|z_t\|) \;\approx\; \mathrm{sim}_{\text{sem}}(\mathrm{patch}_r, \mathrm{patch}_t)$$

This is empirically supported by DINOv2's self-supervised pre-training on 142M images using image-level + patch-level discriminative losses (Oquab et al., 2023). It is *not* guaranteed: distributional shift, polysemy, and visual ambiguity all break it on a non-trivial fraction of patches.

Bidirectional matching exists precisely because this fails on a tail of patches. Cycle-consistency filtering — a non-parametric robustness check — discards the cases where forward similarity does not survive a round-trip back to the reference mask.

Why this matters statistically

If feature similarity was a noisy but unbiased proxy for semantic similarity, raw forward matching would suffice — averaging over enough patches would recover the truth. The empirical evidence (SD ratio 2.46×, 5 catastrophic-failure classes) shows it is **biased on a structured subset** of inputs. Cycle-consistency filtering identifies and removes that biased mass.

Setup & Reproduction

Image encoder	Segmenter	Dataset	Episodes	Hardware	Trainable params
DINOv2 ViT-L/14 (frozen)	SAM ViT-H (frozen)	FSS-1000 fold 0 (20 classes)	120 / class · 2,400 total	NVIDIA L4 Colab Pro	*0*

Fold 0 only due to compute (3–7 GPU-h / fold; full 12-fold sweep = 36–80 h). Paper's per-fold variance $\sim \pm 1$ mIoU bounds the comparability gap.

For each regime ϕ, per-class mean IoU is a sample mean over independent episodes:

PER-CLASS ESTIMATOR

$$\widehat{\mathrm{mIoU}}_c(\phi) = \frac{1}{120}\sum_{i=1}^{120} \mathrm{IoU}_{i,c}$$

Method: Three-Stage Pipeline

Matcher composes DINOv2 (feature matching) and SAM (class-agnostic segmentation) through three training-free stages:

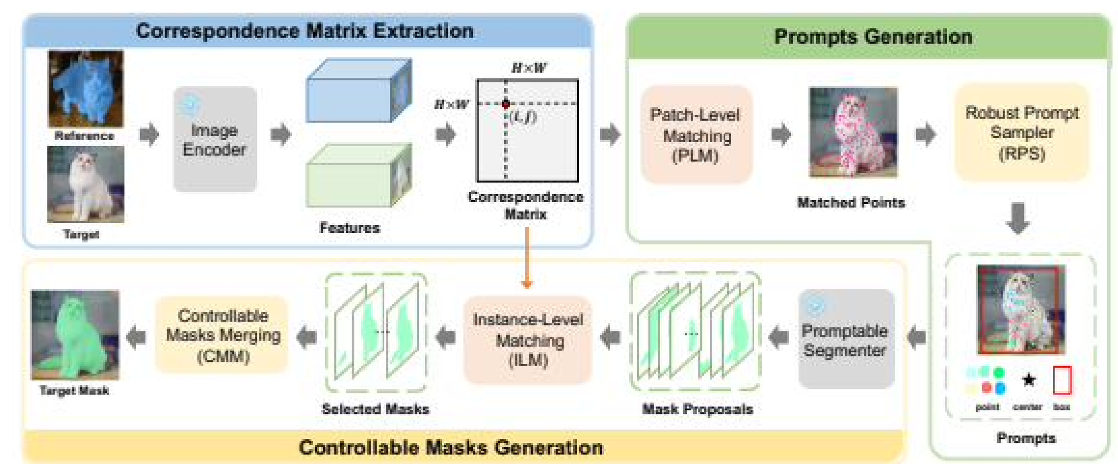

Figure 1: An overview of Matcher. Our training-free framework addresses various segmentation tasks through three operations: Correspondence Matrix Extraction, Prompts Generation, and Controllable Masks Generation.

Step 1. CME (Correspondence Matrix Extraction) — DINOv2 encodes both images; cosine similarity matrix between every pair of patches.

$$S_{ij} = \langle z_r^{(i)}, z_t^{(j)}\rangle \,/\, (\|z_r^{(i)}\|\cdot\|z_t^{(j)}\|)$$

DINOv2 patch features $z_r^{(i)}, z_t^{(j)} \in \mathbb{R}^d$, similarity matrix $S \in \mathbb{R}^{HW\times HW}$.

Step 2. PG (Prompts Generation) — *bidirectional matching* filters outliers via Hungarian round-trip; k-means + clusters retained matches; multi-scale prompts feed SAM.

Forward: assignment $\pi^{\rightarrow}$ minimizes transport cost reference $\rightarrow$ target.

Reverse: $\pi^{\leftarrow}$ sends candidates back. **Filter** retains only points whose round-trip lands in m_r:

OUTLIER FILTER

$$\hat{P} = \{\, p_t^{(i)} \in P_t^{\rightarrow} \;:\; \pi^{\leftarrow}(p_t^{(i)}) \in m_r \,\}$$

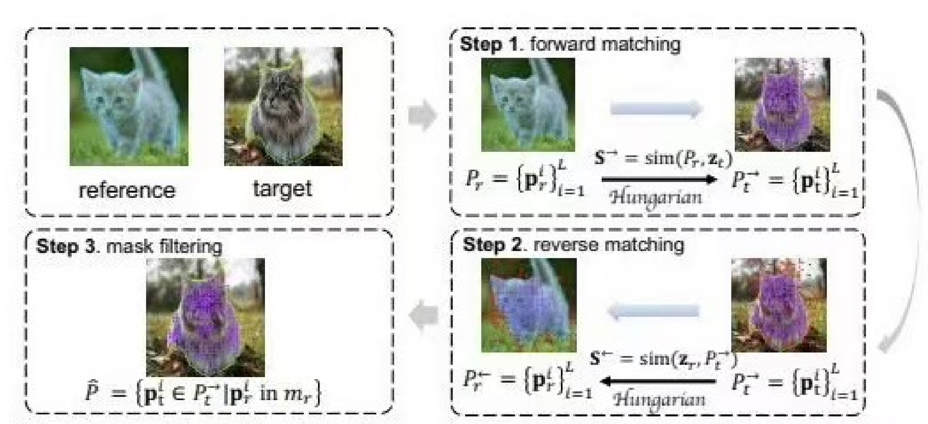

Figure 2: Illustration of the proposed bidirectional matching. Bidirectional matching consists of three steps: forward matching, reverse matching, and mask filtering. Purple points denote the matched points. Red points denote the outliers.

Bidirectional matching is a **cycle-consistency outlier filter**: only matches that survive the round-trip pass through. Equivalent to a non-parametric robustness check on each candidate match.

Step 3. CMG (Controllable Masks Generation) — ILM scores SAM proposals using EMD, purity, coverage; top-k merged into final mask $\hat{m}_t$.

MASK PROPOSAL SCORE

$$\mathrm{score}(m_p) = \alpha(1 - \mathrm{EMD}(m_p, m_r)) + \beta\cdot\mathrm{purity}(m_p)\cdot\mathrm{coverage}(m_p)^{\lambda}$$

EMD = Earth-Mover's Distance over patch features inside m_p, m_r; purity, coverage measure agreement with $\hat{P}$. Top-k merged. $\alpha = 0.8, \beta = 0.2, \lambda = 1.0, k = 10$.

Results: Mean & Distribution

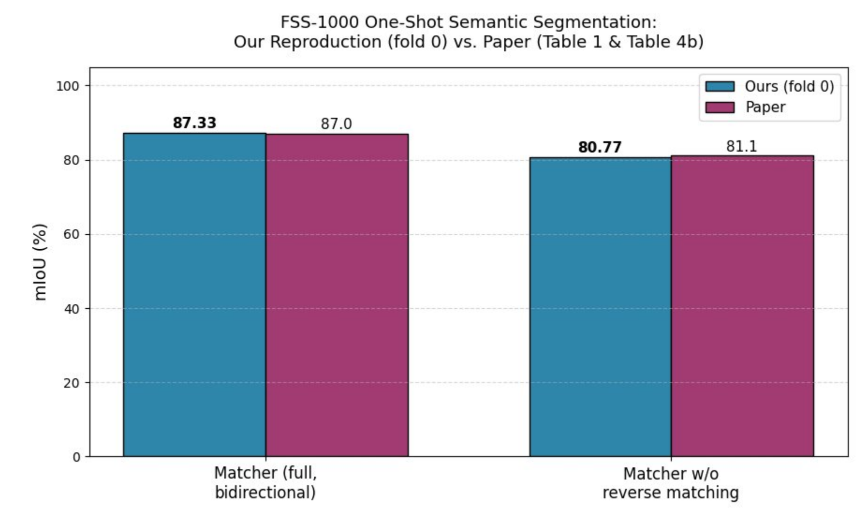

Fig. 3 — *Mean mIoU. Both within 0.33 of paper.*

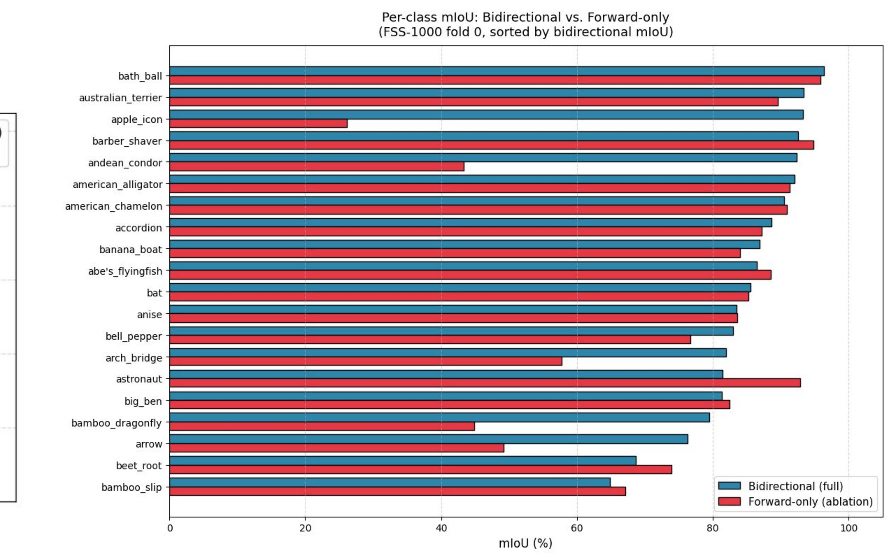

Fig. 4 — *Per-class IoU (20 classes, sorted).*

Setting (paper reference)	Ours	Paper	Δ
Matcher full — paper Table 1, FSS-1000	**87.33**	87.0	+0.33
w/o reverse matching — paper Table 4b	**80.77**	81.1	−0.33
Bidirectional contribution Δ	**+6.56**	**+5.9**	**+0.66**

Distributional view of the ablation

Fig. 3 reports a single point estimate per condition. Fig. 4 disaggregates by class — and reveals that the −6.56 mIoU mean shift is *not* a uniform downward translation. Both agree on most classes, the gap focuses on the left tail of catastrophic failures.

SD per-class IoU (full)	8.06	SD per-class IoU (forward)	19.80
SD ratio (forward / full)	2.46×	Worst-class IoU (full → fwd)	64.9 → 26.1

→ *Removing the cycle-consistency filter* **nearly triples per-class SD**. *Mean shift driven by extreme left-tail mass — a tail-risk story, not a location-shift story.*

Beyond the Paper

(1) Variance, not mean, is the story - Bidirectional matching is a **tail-risk reducer**, not a location shifter. Per-class SD drops 19.80 → 8.06; worst class moves 26.1 → 64.9 IoU. The paper does not quantify this distributional effect.

(2) The filter is not Pareto-improving - **5 of 20 classes are worse** with bidirectional matching, e.g., `astronaut` (−11.4) and `beet_root` (−5.3). This paper does not acknowledge a bias–variance trade-off.

(3) Reproducibility issue in released code - The paper's *Table 4a* "no ILM" condition ($\alpha = \beta = 0$) **cannot be cleanly reproduced**: division by zero in `topk_scores.max()` produces NaN. We document and pivot to paper Table 4b.

(4) Reframing the contribution - Matcher succeeds because of **cycle-consistency outlier filtering**, not because the features alone are enough. The composition fails catastrophically on visually-ambiguous classes without it.

Conclusion

Faithful reproduction ($\Delta < 0.33$ mIoU on Tables 1 and 4b). The substantive finding: **bidirectional matching is best understood as a robust outlier filter, not a feature improvement** *— its primary effect is on per-class IoU variance, a tail-risk reduction the aggregate mean obscures.*

REFERENCES [1] Liu, Zhu, Li, et al. *Matcher: Segment Anything with One-Shot Using All-Purpose Feature Matching.* ICLR 2024. arXiv:2305.13310 [2] Oquab et al. *DINOv2.* arXiv:2304.07193, 2023 [3] Kirillov et al. *Segment Anything.* arXiv:2304.02643, 2023 [4] Li et al. *FSS-1000.* CVPR 2020 `https://github.com/JordanXi6/cs4782-matcher`